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PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

The Detective's Dossier

Team Name : *Team ID :*	*Code Detektives* *DSCPP-III-2026-T027*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Bahuguna, Ashutosh –2510370440 ashutoshbahuguna2468@gmail.com* 
Team member 2 *(Last Name, name: student ID: email, picture):*	*Pundir, Snehal – 2510370590 snehalpundir21052@gmail.com* 
Team member 3 *(Last Name, name: student ID: email, picture):*	*Somraan, Vidhi – 2510380302 Vidhisomraan@gmail.com* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

- Murder mysteries and detective stories have always intrigued us because of the clues, suspects, observations and the intensity of it all.
- Growing up watching shows like CID and always having dreamt of solving a murder mystery added to our motive, and hence we decided to create this project in the form of an interactive game.
- In our project "The Detective's Dossier", players will take on the role of a detective in a 2D murder mystery game.
- They will have to collect clues, visit various locations, examine evidence and question the suspects.
- They will have to connect all the clues to find the culprit.
- Our primary reason to select this project was that it allows a practical use of Data Structures and Algorithms.
- Several DSA concepts can be used in the investigation.

Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

- The project will follow a modern approach, where the game interface, application logic, investigating system and data storage will be separated and planned as independent modules.
- C++ programming language will be used to build the core logic of the game, and Object-Oriented Programming principles of classes, objects and encapsulation will be applied to structure the detectives, suspects, cases, evidence and locations.
- Qt will be used to develop the 2D graphical interface for the game, where players will be able to interact with the game.
- SQLite will be utilized to store the details of the cases and investigation progress locally.
- Data Structures and Algorithms will be the core of the investigation system, where graphs can be used to model locations and clues.
- BFS and DFS algorithms can be used to traverse them.
- Searching and hashing algorithms can be used to quickly find investigation details.

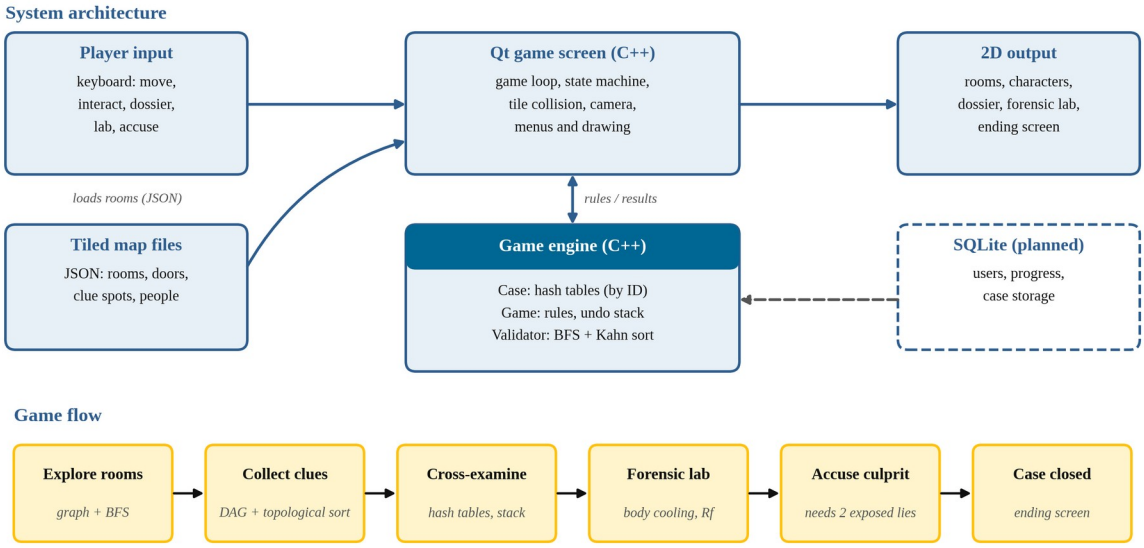


Figure 1: Architecture and game flow of The Detective's Dossier.

Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task Completed	Team Member
Finalized the project idea and identified the main objectives of the detective game.	All team members
Planned the basic project structure and identified the technologies required for development.	Ashutosh Bahuguna
Discussed the game design, locations, clues, suspects and planned features.	Snehal Pundir
Started working on the SQLite database for storing user details and game progress.	Vidhi Somraan

Challenges/Roadblocks (7 pts)

(Describe the challenges that you have faced or are facing so far and how you plan to solve them. Max 300 words).

- One of the main challenges faced while planning the game is that it combines an interesting murder mystery with the practical application of Data Structures and Algorithms.
- The relationships between suspects, clues, evidence and locations need to be organized properly so that the investigation remains logical.
- Another challenge is to select a suitable structure for the C++ application and to decide how different modules will communicate with each other.
- Implementing the concepts of DSA in a meaningful way is also an important part of the project.
- The team needs to determine where graphs, traversal algorithms and searching techniques can be used effectively.
- To address these challenges, the team plans to develop the project in smaller modules, study the required concepts, test individual components and integrate them.

Tasks Pending (7 pts)

(Describe the main tasks that you still need to complete. Max 250 words).

Task Pending	Team Member (to complete the task)
1. Complete the top-down map and export it for integration into the game.	Ashutosh Bahuguna
2. Implement suitable DSA concepts and connect the game modules.	Ashutosh Bahuguna
3. Implement player movement and location interaction.	Snehal Pundir
4. Develop the GUI and test the complete game.	Snehal Pundir
5. Improve the investigation logic, clue collection and suspect questioning features.	Vidhi Somraan
6. Integrate SQLite for storing game progress, if required.	Vidhi Somraan

Project Outcome/Deliverables (2 pts)

(Describe what are the key outcomes / deliverables of the project. Max 200 words).

- The main deliverable of this project is an interactive 2D detective game based on a murder mystery, where players will be able to explore different locations, collect clues, question suspects and make accusations.
- The project aims to apply C++ programming, Object-Oriented Programming, and Data Structures and Algorithms in a practical way.
- A top-down map is being designed using Tiled, and the game will be further developed by integrating the map with the investigation logic.
- The final outcome will be an engaging game that encourages logical reasoning and problem-solving.

Progress Overview (2 pts)

(Summarize how much of the project is done, what's behind schedule, what's ahead of schedule. Max 200 words.)

- The project is currently in its development stage.
- The basic C++ program has been created and successfully executed in VS Code, with options to investigate locations, question suspects, view clues and make an accusation.
- The top-down map for different locations is currently being designed using Tiled.
- Further work is required to integrate the map with the C++ program, improve the gameplay, implement the planned data structures and algorithms concepts, and develop the graphical interface.
- Database integration and testing are also pending.
- The team is working on these tasks step by step.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

- Repository link: https://github.com/AshutoshBahuguna206/PBL-Sem3-DSCPP-

Testing and Validation Status (2 pts)

(Provide information about any tests conducted)

Test Type	Status (Pass/Fail)	Notes
Basic C++ program execution	Pass	The basic program was successfully executed in VS Code.
Game feature testing	Pending	Complete testing will be performed after implementing the remaining features.
Map integration testing	Pending	Testing will be done after integrating the Tiled map with the C++ program.

Deliverables Progress (2 pts)

(Summarize the current status of all key project deliverables mentioned earlier. Indicate whether each deliverable is completed, in progress, or pending.)

- The project is currently in its initial development stage.
- The project concept and basic structure have been planned.
- The basic C++ program has been executed successfully.
- The top-down map for different locations is currently being designed using Tiled.
- The graphical interface, map integration, investigation features and implementation of DSA concepts are still in progress or pending.
- Database integration and complete game testing are also pending.
- The team will work on these deliverables step by step to achieve the project objectives.